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Chip package structure with lid

Abstract

A chip package structure is provided. The chip package structure includes a wiring substrate. The chip package structure includes a first chip structure over the wiring substrate. The chip package structure includes a heat-spreading lid over the wiring substrate and covering the first chip structure. The heat-spreading lid includes a ring structure and a top plate, the ring structure surrounds the first chip structure, the top plate covers the ring structure and the first chip structure, and the first chip structure has a first sidewall and a second sidewall opposite to the first sidewall.

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Background/Summary

CROSS REFERENCE (1) This application is a Divisional of U.S. application Ser. No. 17/459,347, filed on Aug. 27, 2021, now U.S. Pat. No. 12,033,913, the entirety of which is incorporated by reference herein.

BACKGROUND

(1) Semiconductor devices are used in a variety of electronic applications, such as personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating layers or dielectric layers, conductive layers, and semiconductor layers over a semiconductor substrate, and patterning the various material layers **1** using photolithography processes and etching processes to form circuit components and elements thereon.

(2) Many integrated circuits (IC) are typically manufactured on a semiconductor wafer. Technological advances in IC materials and design have produced generations of ICs. Each generation has smaller and more complex circuits than the previous generation. The dies of the wafer may be processed and packaged, and various technologies have been developed for packaging. Since the chip package structure may need to include multiple chips with multiple functions, it is a challenge to form a reliable chip package structure with multiple chips.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. **1A-1D** are cross-sectional views of various stages of a process for forming a chip package structure, in accordance with some embodiments.

(3) FIGS. **1A-1** to **1D-1** are top views of the chip package structure of FIGS. **1A-1D**, in accordance with some embodiments.

(4) FIG. **1A-2** is a cross-sectional view illustrating the chip package structure along a sectional line II-II' in FIG. **1A-1**, in accordance with some embodiments.

(5) FIG. **1B-2** is a cross-sectional view illustrating the chip package structure along a sectional line II-II' in FIG. **1B-1**, in accordance with some embodiments.

(6) FIG. **1D-2** is a cross-sectional view illustrating the chip package structure along a sectional line II-II' in FIG. **1D-1**, in accordance with some embodiments.

(7) FIG. **1D-3** is a perspective view illustrating the chip package structure of FIG. **1D**, in accordance with some embodiments.

(8) FIG. **2A** is a top view of a chip package structure, in accordance with some embodiments.

(9) FIG. **2B** is a cross-sectional view illustrating the chip package structure along a sectional line

2B-2B' in FIG. 2A, in accordance with some embodiments.

(10) FIG. 3A is a top view of a chip package structure, in accordance with some embodiments.

(11) FIG. 3B is a cross-sectional view illustrating the chip package structure along a sectional line 3B-3B' in FIG. 3A, in accordance with some embodiments.

(12) FIG. 4 is a top view of a chip package structure, in accordance with some embodiments.

(13) FIG. 5 is a top view of a chip package structure, in accordance with some embodiments.

(14) FIG. 6 is a top view of a chip package structure, in accordance with some embodiments.

(15) FIG. 7 is a top view of a chip package structure, in accordance with some embodiments.

(16) FIG. 8 is a top view of a chip package structure, in accordance with some embodiments.

(17) FIG. 9 is a top view of a chip package structure, in accordance with some embodiments.

(18) FIG. 10A is a top view of a chip package structure, in accordance with some embodiments.

(19) FIG. 10B is a cross-sectional view illustrating the chip package structure along a sectional line 10B-10B' in FIG. 10A, in accordance with some embodiments.

(20) FIG. 10C is a cross-sectional view illustrating the chip package structure along a sectional line 10C-10C' in FIG. 10A, in accordance with some embodiments.

(21) FIG. 11A is a top view of a chip package structure, in accordance with some embodiments.

(22) FIG. 11B is a cross-sectional view illustrating the chip package structure along a sectional line 11B-11B' in FIG. 11A, in accordance with some embodiments.

(23) FIG. 12A is a top view of a chip package structure, in accordance with some embodiments.

(24) FIG. 12B is a cross-sectional view illustrating the chip package structure along a sectional line 12B-12B' in FIG. 12A, in accordance with some embodiments.

(25) FIGS. 13A-13B are cross-sectional views of various stages of a process for forming a chip package structure, in accordance with some embodiments.

(26) FIGS. 13A-1 to 13B-1 are top views of the chip package structure of FIGS. 13A-13B, in accordance with some embodiments.

(27) FIG. 13B-2 is a cross-sectional view illustrating the chip package structure along a sectional line II-II' in FIG. 13B-1, in accordance with some embodiments.

DETAILED DESCRIPTION

(28) The following disclosure provides many different embodiments, or examples, for implementing different features of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(29) Furthermore, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(30) The term “substantially” in the description, such as in “substantially flat” or in “substantially coplanar”, etc., will be understood by the person skilled in the art. In some embodiments the adjective substantially may be removed. Where applicable, the term “substantially” may also include embodiments with “entirely”, “completely”, “all”, etc. The term “substantially” may be

varied in different technologies and be in the deviation range understood by the skilled in the art. For example, the term “substantially” may also relate to 90% of what is specified or higher, such as 95% of what is specified or higher, especially 99% of what is specified or higher, including 100% of what is specified, though the present invention is not limited thereto. Furthermore, terms such as “substantially parallel” or “substantially perpendicular” may be interpreted as not to exclude insignificant deviation from the specified arrangement and may include for example deviations of up to 10°. The word “substantially” does not exclude “completely” e.g. a composition which is “substantially free” from Y may be completely free from Y.

(31) The term “about” may be varied in different technologies and be in the deviation range understood by the skilled in the art. The term “about” in conjunction with a specific distance or size is to be interpreted so as not to exclude insignificant deviation from the specified distance or size. For example, the term “about” may include deviations of up to 10% of what is specified, though the present invention is not limited thereto. The term “about” in relation to a numerical value x may mean $x \pm 5$ or 10% of what is specified, though the present invention is not limited thereto.

(32) Some embodiments of the disclosure are described. Additional operations can be provided before, during, and/or after the stages described in these embodiments. Some of the stages that are described can be replaced or eliminated for different embodiments. Additional features can be added to the semiconductor device structure. Some of the features described below can be replaced or eliminated for different embodiments. Although some embodiments are discussed with operations performed in a particular order, these operations may be performed in another logical order.

(33) FIGS. 1A-1D are cross-sectional views of various stages of a process for forming a chip package structure, in accordance with some embodiments. FIGS. 1A-1 to 1D-1 are top views of the chip package structure of FIGS. 1A-1D, in accordance with some embodiments. FIGS. 1A-1D are cross-sectional views illustrating the chip package structure along a sectional line I-I' in FIGS. 1A-1 to 1D-1, in accordance with some embodiments.

(34) As shown in FIGS. 1A and 1A-1, a wiring substrate **110** is provided, in accordance with some embodiments. The wiring substrate **110** includes a dielectric structure **112**, wiring layers **114**, and conductive vias **116**, in accordance with some embodiments. The wiring layers **114** and the conductive vias **116** are formed in the dielectric structure **112**, in accordance with some embodiments.

(35) The conductive vias **116** are electrically connected between different wiring layers **114**, in accordance with some embodiments. For the sake of simplicity, FIG. 1A only shows two of the wiring layers **114**, in accordance with some embodiments.

(36) The dielectric structure **112** is made of an insulating material such as a polymer material (e.g., polybenzoxazole, polyimide, or a photosensitive material), nitride (e.g., silicon nitride), oxide (e.g., silicon oxide), silicon oxynitride, or the like, in accordance with some embodiments. The dielectric structure **112** is formed using deposition processes (e.g. chemical vapor deposition processes or physical vapor deposition processes), photolithography processes, and etching processes, in accordance with some embodiments.

(37) The wiring layers **114** are made of a conductive material, such as metal (e.g. copper, aluminum, or tungsten) or alloys thereof, in accordance with some embodiments. The conductive vias **116** are made of a conductive material, such as metal (e.g. copper, aluminum, or tungsten) or alloys thereof, in accordance with some embodiments.

(38) In some embodiments, the wiring substrate **110** includes conductive pads (not shown). The conductive pads are formed over the dielectric structure **112**, in accordance with some embodiments. The conductive vias **116** are electrically connected between the wiring layer **114** and the conductive pads, in accordance with some embodiments.

(39) The conductive pads are made of a conductive material, such as metal (e.g. copper, aluminum, or tungsten) or alloys thereof, in accordance with some embodiments. In some embodiments, the

wiring layers **114**, the conductive vias **116**, and the conductive pads are made of the same material. In some other embodiments, the wiring layers **114**, the conductive vias **116**, and the conductive pads are made of different materials.

(40) As shown in FIGS. **1A** and **1A-1**, chip structures **122**, **122'**, and **132** are bonded to the wiring substrate **110** through conductive connectors **124** and **134**, in accordance with some embodiments. The chip structures **122**, **122'**, and **132** are spaced apart from each other by gaps **G10** and **G11**, in accordance with some embodiments.

(41) The chip structure **132** has sidewalls **132a**, **132b**, **132c**, and **132d**, in accordance with some embodiments. The chip structure **132** has corners **C1** and **C2**, in accordance with some embodiments. The corner **C1** is between the sidewalls **132a** and **132c**, in accordance with some embodiments. The corner **C2** is between the sidewalls **132a** and **132d**, in accordance with some embodiments.

(42) The chip structure **122** has sidewalls **122a**, **122b**, **122c**, and **122d**, in accordance with some embodiments. The chip structure **122** has a corner **C3** between the sidewalls **122a** and **122d**, in accordance with some embodiments. The chip structure **122'** has sidewalls **122a'**, **122b'**, **122c'**, and **122d'**, in accordance with some embodiments. The chip structure **122'** has a corner **C4** between the sidewalls **122a'** and **122c'**, in accordance with some embodiments.

(43) The chip structures **122**, **122'**, and **132** are chips and/or chip packages, in accordance with some embodiments. In some embodiments, the chip structures **122** and **122'** are chip packages, such as dynamic random access memory (DRAM) packages. The chip packages include chip scale packages, such as wafer level chip scale packages. In some embodiments, each chip package includes one chip. In some other embodiments, each chip package includes multiple chips, which are arranged side by side or stacked with each other (e.g., a 3D packaging or a 3DIC device). In some embodiments, the chip structure **132** is a chip, such as a central processing unit (CPU) chip.

(44) The chip of the chip structures **122**, **122'**, and **132** includes a substrate, in accordance with some embodiments. In some embodiments, the substrate is made of an elementary semiconductor material including silicon or germanium in a single crystal structure, a polycrystal structure, or an amorphous structure.

(45) In some other embodiments, the substrate is made of a compound semiconductor, such as silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, an alloy semiconductor, such as SiGe, or GaAsP, or a combination thereof. The substrate may also include multi-layer semiconductors, semiconductor on insulator (SOI) (such as silicon on insulator or germanium on insulator), or a combination thereof.

(46) In some embodiments, the substrate includes various device elements. In some embodiments, the various device elements are formed in and/or over the substrate. The device elements are not shown in figures for the purpose of simplicity and clarity. Examples of the various device elements include active devices, passive devices, other suitable elements, or a combination thereof. The active devices may include transistors or diodes (not shown) formed at a surface of the substrate. The passive devices include resistors, capacitors, or other suitable passive devices.

(47) For example, the transistors may be metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high-voltage transistors, high-frequency transistors, p-channel and/or n-channel field effect transistors (PFETs/NFETs), etc. Various processes, such as front-end-of-line (FEOL) semiconductor fabrication processes, are performed to form the various device elements. The FEOL semiconductor fabrication processes may include deposition, etching, implantation, photolithography, annealing, planarization, one or more other applicable processes, or a combination thereof.

(48) In some embodiments, isolation features (not shown) are formed in the substrate. The isolation features are used to define active regions and electrically isolate various device elements formed in and/or over the substrate in the active regions. In some embodiments, the isolation features include

shallow trench isolation (STI) features, local oxidation of silicon (LOCOS) features, other suitable isolation features, or a combination thereof.

(49) The conductive connectors **124** are physically and electrically connected between the chip structure **122** or **122'** and the wiring substrate **110**, in accordance with some embodiments. Each conductive connector **124** includes, for example, a conductive pillar **124a** and a solder bump **124b**, in accordance with some embodiments.

(50) Each conductive pillar **124a** is connected to the chip structure **122** or **122'**, in accordance with some embodiments. Each solder bump **124b** is connected between the corresponding conductive pillar **124a** and the wiring substrate **110**, in accordance with some embodiments.

(51) The conductive pillars **124a** are made of a conductive material such as copper (Cu), aluminum (Al), tungsten (W), cobalt (Co), nickel (Ni), the like, or a combination thereof, in accordance with some embodiments. The conductive pillars **124a** are formed using a plating process such as an electroplating process, in accordance with some embodiments. The solder bumps **124b** are made of a conductive material such as tin (Sn), the like, or alloys thereof, in accordance with some embodiments.

(52) The conductive connectors **134** are physically and electrically connected between the chip structure **132** and the wiring substrate **110**, in accordance with some embodiments. Each conductive connector **134** includes, for example, a conductive pillar **134a** and a solder bump **134b**, in accordance with some embodiments. Each conductive pillar **134a** is connected to the chip structure **132**, in accordance with some embodiments. Each solder bump **134b** is connected between the conductive pillar **134a** and the wiring substrate **110**, in accordance with some embodiments.

(53) The conductive pillars **134a** are made of a conductive material such as copper (Cu), aluminum (Al), tungsten (W), cobalt (Co), nickel (Ni), the like, or a combination thereof, in accordance with some embodiments. The conductive pillars **134a** are formed using a plating process such as an electroplating process, in accordance with some embodiments. The solder bumps **134b** are made of a conductive material such as tin (Sn), the like, or alloys thereof, in accordance with some embodiments.

(54) FIG. **1A-2** is a cross-sectional view illustrating the chip package structure along a sectional line II-II' in FIG. **1A-1**, in accordance with some embodiments. As shown in FIGS. **1A-1** and **1A-2**, devices **140** are bonded to the wiring substrate **110** by, for example, surface mount technology (SMT), in accordance with some embodiments. The devices **140** include passive devices, other suitable devices, or combinations thereof. The passive devices include resistors, capacitors, inductors, or other suitable passive devices.

(55) As shown in FIG. **1A**, an underfill layer **142** is formed between the chip structures **122** and **122'** and the wiring substrate **110**, in accordance with some embodiments. The underfill layer **142** surrounds the conductive connectors **124** and the chip structures **122** and **122'**, in accordance with some embodiments. The underfill layer **142** is made of an insulating material, such as a polymer material, in accordance with some embodiments.

(56) As shown in FIGS. **1A** and **1A-2**, an underfill layer **144** is formed between the chip structure **132** and the wiring substrate **110**, in accordance with some embodiments. The underfill layer **144** surrounds the conductive connectors **134** and the chip structure **132**, in accordance with some embodiments. The underfill layer **144** is made of an insulating material, such as a polymer material, in accordance with some embodiments.

(57) For the sake of simplicity, FIG. **1A-1** omits depicting the underfill layers **142** and **144**, in accordance with some embodiments. Similarly, FIGS. **1B-1** to **1D-1**, **2A**, **3A**, **4-9**, **10A**, **11A**, **12A**, **13A-1**, and **13B-1** omit depicting the underfill layers **142** and **144**, in accordance with some embodiments.

(58) As shown in FIGS. **1A** and **1A-1**, an adhesive layer **150** is formed over the wiring substrate **110**, in accordance with some embodiments. As shown in FIG. **1A-1**, the adhesive layer **150** has a

ring shape, in accordance with some embodiments. The adhesive layer **150** has an opening **152**, in accordance with some embodiments. The chip structures **122**, **122'**, and **132** are in the opening **152**, in accordance with some embodiments.

(59) The opening **152** has an inner wall **152a**, in accordance with some embodiments. The inner wall **152a** has a recess **152b**, in accordance with some embodiments. The recess **152b** faces the chip structure **132**, in accordance with some embodiments. The recess **152b** faces the corners **C1** and **C2** of the chip structure **132**, the corner **C3** of the chip structure **122**, and the corner **C4** of the chip structure **122'**, in accordance with some embodiments.

(60) As shown in FIGS. **1A-1** and **1A-2**, the sidewall **132a** of the chip structure **132** is spaced apart from the adhesive layer **150** by a gap **G1**, in accordance with some embodiments. The sidewall **132b** of the chip structure **132** is spaced apart from the adhesive layer **150** by a gap **G2**, in accordance with some embodiments.

(61) The gap **G1** is narrower than the gap **G2**, in accordance with some embodiments. That is, a distance **D1** between the sidewall **132a** and the adhesive layer **150** is less than a distance **D2** between the sidewall **132b** and the adhesive layer **150**, in accordance with some embodiments. The adhesive layer **150** is made of a polymer (e.g., epoxy or silicone) or a combination of polymer and metal (e.g., a silver paste), in accordance with some embodiments.

(62) As shown in FIGS. **1B** and **1B-1**, a ring structure **160** is disposed over the adhesive layer **150**, in accordance with some embodiments. The adhesive layer **150** is adhered between the ring structure **160** and the wiring substrate **110**, in accordance with some embodiments. The ring structure **160** has an opening **162**, in accordance with some embodiments. The chip structures **122**, **122'**, and **132** are in the opening **162**, in accordance with some embodiments. The ring structure **160** surrounds the chip structures **122**, **122'**, and **132**, in accordance with some embodiments.

(63) The opening **162** has an inner wall **162a**, in accordance with some embodiments. The inner wall **162a** has a recess **162b**, in accordance with some embodiments. The recess **162b** faces the chip structure **132**, in accordance with some embodiments. The recess **162b** faces the corners **C1** and **C2** of the chip structure **132**, the corner **C3** of the chip structure **122**, and the corner **C4** of the chip structure **122'**, in accordance with some embodiments.

(64) FIG. **1B-2** is a cross-sectional view illustrating the chip package structure along a sectional line II-II' in FIG. **1B-1**, in accordance with some embodiments. As shown in FIGS. **1B-1** and **1B-2**, the ring structure **160** has portions **164** and **166**, in accordance with some embodiments. In some embodiments, a line width **W166** of the portion **166** is less than a line width **W164** of the portion **164**.

(65) The portion **164** is also referred to as a wide portion, and the portion **166** is also referred to as a narrow portion, in accordance with some embodiments. The chip structures **122**, **122'**, and **132** are between the portions **164** and **166**, in accordance with some embodiments.

(66) The sidewall **132a** of the chip structure **132** is spaced apart from the portion **166** by a gap **G3**, in accordance with some embodiments. The sidewall **132b** of the chip structure **132** is spaced apart from the portion **164** by a gap **G4**, in accordance with some embodiments. The gap **G3** is narrower than the gap **G4**, in accordance with some embodiments. That is, the portion **166** is closer to the chip structure **132** than the portion **164**, in accordance with some embodiments. In some embodiments, a distance **D3** between the sidewall **132a** and the portion **166** is less than a distance **D4** between the sidewall **132b** and the portion **164**, in accordance with some embodiments. In some embodiments, a difference between the distances **D4** and **D3** ranges from about 1 μm to about 5500 μm .

(67) As shown in FIG. **1B-1**, the sidewall **122a** of the chip structure **122** is spaced apart from the ring structure **160** by a gap **G5**, in accordance with some embodiments. The sidewall **122b** of the chip structure **122** is spaced apart from the ring structure **160** by a gap **G6**, in accordance with some embodiments. The gaps **G5** and **G6** have substantially the same width, in accordance with some embodiments. That is, a distance **D5** between the sidewall **122a** and the ring structure **160** is

substantially equal to a distance **D6** between the sidewall **122b** and the ring structure **160**, in accordance with some embodiments.

(68) Similarly, as shown in FIG. **1B-1**, the sidewall **122a'** of the chip structure **122'** is spaced apart from the ring structure **160** by a gap **G5'**, in accordance with some embodiments. The sidewall **122b'** of the chip structure **122'** is spaced apart from the ring structure **160** by a gap **G6'**, in accordance with some embodiments. The gaps **G5'** and **G6'** have substantially the same width, in accordance with some embodiments. That is, a distance **D5'** between the sidewall **122a'** and the ring structure **160** is substantially equal to a distance **D6'** between the sidewall **122b'** and the ring structure **160**, in accordance with some embodiments.

(69) The ring structure **160** is made of a rigid material, such as metal (e.g., copper or iron), alloys thereof (e.g., stainless steel), or another suitable material which is more rigid than the wiring substrate **110**, in accordance with some embodiments.

(70) As shown in FIGS. **1C** and **1C-1**, an adhesive layer **172** is formed over the ring structure **160**, in accordance with some embodiments. The adhesive layer **172** is made of a polymer (e.g., epoxy or silicone) or a combination of polymer and metal (e.g., a silver paste), in accordance with some embodiments.

(71) As shown in FIGS. **1C** and **1C-1**, an adhesive layer **174** is formed over the chip structure **132**, in accordance with some embodiments. The adhesive layer **174** is made of a combination of polymer and metal (e.g., a silver paste) or a polymer (e.g., epoxy or silicone), in accordance with some embodiments. In some embodiments, the adhesive layers **172** and **174** are made of different materials. In some other embodiments, the adhesive layers **172** and **174** are made of the same material.

(72) As shown in FIGS. **1D** and **1D-1**, a top plate **180** is disposed over the adhesive layers **172** and **174**, in accordance with some embodiments. The top plate **180**, the adhesive layer **172**, and the ring structure **160** together form a heat-spreading lid **190**, in accordance with some embodiments. In this step, a chip package structure **100** is substantially formed, in accordance with some embodiments.

(73) The adhesive layer **172** is adhered between the ring structure **160** and the top plate **180**, in accordance with some embodiments. The adhesive layer **174** is adhered between the chip structure **132** and the top plate **180**, in accordance with some embodiments. The top plate **180** covers the ring structure **160** and the chip structure **132**, in accordance with some embodiments.

(74) The top plate **180** has openings **182** and **184**, in accordance with some embodiments. The openings **182** expose the chip structures **122** and **122'** respectively, in accordance with some embodiments. FIG. **1D-2** is a cross-sectional view illustrating the chip package structure along a sectional line II-II' in FIG. **1D-1**, in accordance with some embodiments. As shown in FIGS. **1D-1** and **1D-2**, the opening **184** is over the gap **G3**, which is between the sidewall **132a** of the chip structure **132** and the portion **166**, in accordance with some embodiments.

(75) As shown in FIG. **1D-1**, a first portion of the opening **184** is between the portion **166** and the chip structure **132**, in accordance with some embodiments. In some embodiments, a second portion of the opening **184** is between the portion **166** and the gap **G10**, which is between the chip structures **122** and **132**. In some embodiments, a third portion of the opening **184** is between the portion **166** and the chip structure **122**.

(76) In some embodiments, a fourth portion of the opening **184** is between the portion **166** and the gap **G11**, which is between the chip structures **122'** and **132**. In some embodiments, a fifth portion of the opening **184** is between the portion **166** and the chip structure **122'**. The opening **184** is between the portion **166** and the corners **C1**, **C2**, **C3** and **C4**, in accordance with some embodiments.

(77) As shown in FIGS. **1D** and **1D-1**, a width **W184** of the opening **184** is greater than a sum of a width **D10** of the gap **G10**, a width **D11** of the gap **G11**, and a width **W132** of the chip structure **132**, in accordance with some embodiments. In some embodiments, a length **L184** of the opening

184 is less than a length **L132** of the chip structure **132**. The width **W184** is greater than the length **L184**, in accordance with some embodiments. The width **W184** is greater than a distance **D9** between the opening **184** and a sidewall **186** of the top plate **180**, in accordance with some embodiments.

(78) As shown in FIG. 1D-1, the gaps **G10** and **G11** extend toward the opening **184**, in accordance with some embodiments. In some embodiments, the opening **184** has a strip shape. The opening **184** has inner walls **184a** and **184b** facing each other, in accordance with some embodiments. The inner wall **184b** is between the inner wall **184a** and the sidewall **132a** of the chip structure **132**, in accordance with some embodiments. The inner walls **184a** and **184b** are both between the sidewall **132a** and the ring structure **160**, in accordance with some embodiments.

(79) FIG. 1D-3 is a perspective view illustrating the chip package structure of FIG. 1D, in accordance with some embodiments. As shown in FIGS. 1D-1, 1D-2, and 1D-3, the opening **184** does not overlap with the chip structure **122**, **122'**, and **132**, in accordance with some embodiments. That is, the opening **184** does not expose the chip structure **122**, **122'**, and **132**, in accordance with some embodiments. In some embodiments, the inner walls **184a**, **172a1**, **162a**, and **152a** of the openings **184**, **172a**, **162**, and **152** of the top plate **180**, the adhesive layer **172**, the ring structure **160**, and the adhesive layer **150** are substantially aligned with or level with or coplanar to each other.

(80) The top plate **180** is made of a rigid material, such as metal (e.g., copper or iron), alloys thereof (e.g., stainless steel), or another suitable material which is more rigid than the wiring substrate **110**, in accordance with some embodiments.

(81) The coefficient of thermal expansion of the chip structures **122**, **122'**, and **132** ranges from about 1 ppm/° C. to about 5 ppm/° C., in accordance with some embodiments. The coefficient of thermal expansion of the heat-spreading lid **190** ranges from about 12 ppm/° C. to about 22 ppm/° C., in accordance with some embodiments.

(82) The coefficient of thermal expansion of the wiring substrate **110** ranges from about 10 ppm/° C. to about 20 ppm/° C., in accordance with some embodiments. The difference between the coefficients of thermal expansion of the chip structures **122**, **122'**, and **132** and the heat-spreading lid **190** (or the wiring substrate **110**) is large, in accordance with some embodiments.

(83) The coefficient of thermal expansion mismatch between the chip structures **122**, **122'**, and **132** and the heat-spreading lid **190** (or the wiring substrate **110**) tends to result in a thermal stress, in accordance with some embodiments. Since the portion **166** (of the ring structure **160**) is closer to the chip structure **132** than the portion **164**, the thermal stress in the adhesive layers **150** and **172** close to the portion **166** is greater than the thermal stress in the adhesive layers **150** and **172** close to the portion **164**, in accordance with some embodiments. Therefore, the thermal stress tends to induce cracks in the adhesive layers **150** and **172** close to the portion **166** and delamination between the adhesive layers **150** and **172** close to the portion **166** and the elements adjacent thereto (e.g., the wiring substrate **110**, the ring structure **160**, and the top plate **180**), in accordance with some embodiments.

(84) Since the thermal stress is transmitted from the chip structure **132** to the portion **166** through the top plate **180** and the wiring substrate **110**, the formation of the opening **184** in the top plate **180** between the chip structure **132** and the portion **166** is able to hinder the transmission of the thermal stress, in accordance with some embodiments. The opening **184** is also able to release the thermal stress, in accordance with some embodiments. Therefore, the opening **184** reduces the thermal stress in the adhesive layers **150** and **172** close to the portion **166**, in accordance with some embodiments.

(85) As a result, the opening **184** prevents the adhesive layers **150** and **172** close to the portion **166** from delamination and developing cracks in subsequent thermal processes, in accordance with some embodiments. Therefore, the opening **184** improves the reliability of the adhesive layers **150** and **172**, in accordance with some embodiments.

(86) In some embodiments, the length **L184** of the opening **184** is greater than the distance **D9** between the opening **184** and the sidewall **186** of the top plate **180**. If the length **L184** is greater than the distance **D9**, the transmission path length of the thermal stress from the corner **C1** or **C2** to the portion **166** of the ring structure **160** is increased, which hinders the transmission of the thermal stress from the corners **C1** and **C2** to the portion **166**, in accordance with some embodiments.

(87) In some embodiments, the length **L184** of the opening **184** is equal to or less than the distance **D9** between the opening **184** and the sidewall **186** of the top plate **180**. If the length **L184** is equal to or less than the distance **D9**, the design of the opening **184** is able to retain more heat dissipation area of the top plate **180** close to the chip structure **132**, in accordance with some embodiments.

(88) As shown in FIGS. **1D-1** and **1D-2**, the formation of the recess **162b** of the ring structure **160** is able to enlarge the distance **D3** between the chip structure **132** and the portion **166** of the ring structure **160**, which reduces the thermal stress in the adhesive layers **150** and **172** close to the portion **166**, in accordance with some embodiments.

(89) In some embodiments, the great thermal stress is at the corners (e.g., **C1** and **C2**) of the chip structure **132**. The opening **184** is between the corners **C1** and **C2** and the portion **166** of the ring structure **160** to hinder the transmission of the thermal stress from the corners **C1** and **C2** to the portion **166**, in accordance with some embodiments.

(90) FIG. **2A** is a top view of a chip package structure **200**, in accordance with some embodiments. FIG. **2B** is a cross-sectional view illustrating the chip package structure **200** along a sectional line **2B-2B'** in FIG. **2A**, in accordance with some embodiments.

(91) As shown in FIGS. **2A** and **2B**, the chip package structure **200** is similar to the chip package structure **100** of FIGS. **1D-1** and **1D-2**, except that the inner wall **184a** of the opening **184** of the top plate **180** is not aligned with the inner walls **172a1**, **162a**, and **152a** of the openings **172a**, **162**, and **152** of the adhesive layer **172**, the ring structure **160**, and the adhesive layer **150**, in accordance with some embodiments.

(92) FIG. **3A** is a top view of a chip package structure **300**, in accordance with some embodiments. FIG. **3B** is a cross-sectional view illustrating the chip package structure **300** along a sectional line **3B-3B'** in FIG. **3A**, in accordance with some embodiments.

(93) As shown in FIGS. **3A** and **3B**, the chip package structure **300** is similar to the chip package structure **100** of FIGS. **1D-1** and **1D-2**, except that the inner wall **184a** of the opening **184** of the top plate **180** is substantially aligned with or level with or coplanar to sidewalls **174a** and **132a** of the adhesive layer **174** and the chip structure **132**, in accordance with some embodiments.

(94) FIG. **4** is a top view of a chip package structure **400**, in accordance with some embodiments. As shown in FIG. **4**, the chip package structure **400** is similar to the chip package structure **100** of FIG. **1D-1**, except that the top plate **180** of the chip package structure **400** has openings **184c** and **184d** spaced apart from each other, in accordance with some embodiments. The top plate **180** of the chip package structure **400** does not have the opening **184** of the top plate **180** of the chip package structure **100** of FIG. **1D-1**, in accordance with some embodiments.

(95) The opening **184c** is between the corner **C1** and the portion **166** of the ring structure **160**, in accordance with some embodiments. The opening **184c** has opposite inner walls **184c1** and **184c2**, in accordance with some embodiments. In some embodiments, a lateral distance **D12** between the sidewall **132c** of the chip structure **132** and the inner wall **184c1** is substantially equal to a lateral distance **D13** between the sidewall **132c** and the inner wall **184c2**.

(96) The opening **184c** is able to hinder the transmission of the thermal stress from the corner **C1** to the portion **166**, in accordance with some embodiments. In some embodiments, the lateral distance **D12** or **D13** is greater than a distance **D16** between the opening **184c** and the sidewall **186** of the top plate **180**. In some embodiments, a length **L184c** of the opening **184c** is greater than the distance **D16**. If the distance **D12** or **D13** or the length **L184c** is greater than the distance **D16**, the transmission path length of the thermal stress from the corner **C1** to the portion **166** of the ring structure **160** is increased, which further hinders the transmission of the thermal stress from the

corner C1 to the portion **166**, in accordance with some embodiments.

(97) In some embodiments, the lateral distance D12 or D13 is equal to or less than the distance D16. In some embodiments, the length L184c is equal to or less than the distance D16. If the distance D12 or D13 or the length L184c is equal to or less than the distance D16, the design of the opening **184c** is able to retain more heat dissipation area of the top plate **180** close to the chip structure **132**, in accordance with some embodiments. In some embodiments, the opening **184c** is between the corner C3 of the chip structure **122** and the portion **166** of the ring structure **160**.

(98) The opening **184d** is between the corner C2 and the portion **166** of the ring structure **160**, in accordance with some embodiments. The opening **184d** has opposite inner walls **184d1** and **184d2**, in accordance with some embodiments. In some embodiments, a lateral distance D14 between the sidewall **132d** of the chip structure **132** and the inner wall **184d1** is substantially equal to a lateral distance D15 between the sidewall **132d** and the inner wall **184d2**.

(99) The opening **184d** is able to hinder the transmission of the thermal stress from the corner C2 to the portion **166**, in accordance with some embodiments. In some embodiments, the lateral distance D14 or D15 is greater than a distance D17 between the opening **184d** and the sidewall **186** of the top plate **180**. In some embodiments, a length L184d of the opening **184d** is greater than the distance D17. If the distance D14 or D15 or the length L184d is greater than the distance D17, the transmission path length of the thermal stress from the corner C2 to the portion **166** of the ring structure **160** is increased, which further hinders the transmission of the thermal stress from the corner C2 to the portion **166**, in accordance with some embodiments.

(100) In some embodiments, the lateral distance D14 or D15 is equal to or less than the distance D17. In some embodiments, the length L184d of the opening **184d** is equal to or less than the distance D17. If the distance D14 or D15 or the length L184d is equal to or less than the distance D17, the design of the opening **184d** is able to retain more heat dissipation area of the top plate **180** close to the chip structure **132**, in accordance with some embodiments. In some embodiments, the opening **184d** is between the corner C4 of the chip structure **122'** and the portion **166** of the ring structure **160**.

(101) FIG. 5 is a top view of a chip package structure **500**, in accordance with some embodiments. As shown in FIG. 5, the chip package structure **500** is similar to the chip package structure **400** of FIG. 4, except that the top plate **180** of the chip package structure **500** further has an opening **184e** between the openings **184c** and **184d**, in accordance with some embodiments.

(102) The openings **184c**, **184d**, and **184e** are spaced apart from each other, in accordance with some embodiments. The opening **184e** is between the sidewall **132a** of the chip structure **132** and the portion **166** of the ring structure **160**, in accordance with some embodiments. The opening **184e** is able to hinder the transmission of the thermal stress from the sidewall **132a** to the portion **166**, in accordance with some embodiments.

(103) FIG. 6 is a top view of a chip package structure **600**, in accordance with some embodiments. As shown in FIG. 6, the chip package structure **600** is similar to the chip package structure **100** of FIG. 1D-1, except that the opening **184** of the top plate **180** of the chip package structure **600** has a U-shape, in accordance with some embodiments.

(104) Specifically, the opening **184** has wide portions **184w1** and **184w2** and a narrow portion **184n**, in accordance with some embodiments. The narrow portion **184n** is connected between the wide portions **184w1** and **184w2**, in accordance with some embodiments. The wide portion **184w1** or **184w2** is wider than the narrow portion **184n**, in accordance with some embodiments. That is, a width W1 of the wide portion **184w1** or **184w2** is greater than a width W2 of the narrow portion **184n**, in accordance with some embodiments.

(105) The wide portion **184w1** is closer to the corner C1 of the chip structure **132** than the narrow portion **184n**, in accordance with some embodiments. The wide portion **184w1** is between the corner C1 and the portion **166** of the ring structure **160**, in accordance with some embodiments. The wide portion **184w2** is closer to the corner C2 of the chip structure **132** than the narrow portion

184n, in accordance with some embodiments. The wide portion **184w2** is between the corner **C2** and the portion **166** of the ring structure **160**, in accordance with some embodiments.

(106) Since the width **W1** of the wide portion **184w1** or **184w2** is large, the transmission path length of the thermal stress from the corner **C1** or **C2** to the portion **166** of the ring structure **160** is increased, which hinders the transmission of the thermal stress from the corners **C1** and **C2** to the portion **166**, in accordance with some embodiments. The design of the narrow portion **184** is able to retain more heat dissipation area of the top plate **180** close to the chip structure **132**, in accordance with some embodiments.

(107) FIG. 7 is a top view of a chip package structure **700**, in accordance with some embodiments. As shown in FIG. 7, the chip package structure **700** is similar to the chip package structure **400** of FIG. 4, except that the openings **184c** and **184d** of the top plate **180** of the chip package structure **700** has an L-shape.

(108) FIG. 8 is a top view of a chip package structure **800**, in accordance with some embodiments. As shown in FIG. 8, the chip package structure **800** is similar to the chip package structure **600** of FIG. 6, except that the opening **184** of the top plate **180** of the chip package structure **800** further has narrow portions **184n1** and **184n2**, in accordance with some embodiments. The wide portion **184w1** is connected between the narrow portions **184n** and **184n1**, in accordance with some embodiments. The wide portion **184w2** is connected between the narrow portions **184n** and **184n2**, in accordance with some embodiments.

(109) FIG. 9 is a top view of a chip package structure **900**, in accordance with some embodiments. As shown in FIG. 9, the chip package structure **900** is similar to the chip package structure **400** of FIG. 4, except that the openings **184c** and **184d** of the top plate **180** of the chip package structure **900** has a T-shape (e.g., an inverted T-shape).

(110) FIG. 10A is a top view of a chip package structure **1000**, in accordance with some embodiments. FIG. 10B is a cross-sectional view illustrating the chip package structure **1000** along a sectional line **10B-10B'** in FIG. 10A, in accordance with some embodiments. FIG. 10C is a cross-sectional view illustrating the chip package structure **1000** along a sectional line **10C-10C'** in FIG. 10A, in accordance with some embodiments.

(111) As shown in FIGS. 10A, 10B, and 10C, the chip package structure **1000** is similar to the chip package structure **100** of FIGS. 1D and 1D-1, except that the chip package structure **1000** further includes chip structures **122A** and **122B**, in accordance with some embodiments. The chip structure **132** is between the chip structures **122A** and **122B**, in accordance with some embodiments.

(112) The chip structures **122A** and **122B** are bonded to the wiring substrate **110** through conductive connectors **124**, in accordance with some embodiments. In some embodiments, an underfill layer **142** is formed between the wiring substrate **110** and the chip structures **122A** and **122B**.

(113) The structures and the materials of the chip structures **122A** and **122B**, the conductive connectors **124**, and the underfill layer **142** are the same as or similar to that of the chip structures **122** and **122'**, the conductive connectors **124**, and the underfill layer **142** of the chip package structure **100** of FIGS. 1D and 1D-1, in accordance with some embodiments.

(114) FIG. 11A is a top view of a chip package structure **1100**, in accordance with some embodiments. FIG. 11B is a cross-sectional view illustrating the chip package structure **1100** along a sectional line **11B-11B'** in FIG. 11A, in accordance with some embodiments.

(115) As shown in FIGS. 11A and 11B, the chip package structure **1100** is similar to the chip package structure **600** of FIG. 6, except that the top plate **180** of the chip package structure **1100** covers the chip structures **122** and **122'**, in accordance with some embodiments. That is, the top plate **180** does not have the openings **182** of the top plate **180** of the chip package structure **600** of FIG. 6, in accordance with some embodiments.

(116) The chip package structure **1100** further has adhesive layers **176** respectively between the chip structure **122** and the top plate **180** and between the chip structure **122'** and the top plate **180**,

in accordance with some embodiments. The materials of the adhesive layers **176** are the same as or similar to that of the adhesive layers **174**, in accordance with some embodiments. The opening **184** surrounds the corners **C1** and **C2**, in accordance with some embodiments.

(117) FIG. **12A** is a top view of a chip package structure **1200**, in accordance with some embodiments. FIG. **12B** is a cross-sectional view illustrating the chip package structure **1200** along a sectional line **12B-12B'** in FIG. **12A**, in accordance with some embodiments.

(118) As shown in FIGS. **12A** and **12B**, the chip package structure **1200** is similar to the chip package structure **1100** of FIG. **11A**, except that the top plate **180** has openings **184c** and **184d** spaced apart from each other, in accordance with some embodiments. The top plate **180** of the chip package structure **1200** does not have the opening **184** of the top plate **180** of the chip package structure **1100** of FIG. **11**, in accordance with some embodiments. The opening **184c** surrounds the corner **C1** of the chip structure **132**, in accordance with some embodiments. The opening **184d** surrounds the corner **C2** of the chip structure **132**, in accordance with some embodiments.

(119) FIGS. **13A-13B** are cross-sectional views of various stages of a process for forming a chip package structure, in accordance with some embodiments. FIGS. **13A-1** to **13B-1** are top views of the chip package structure of FIGS. **13A-13B**, in accordance with some embodiments. FIG. **13B-2** is a cross-sectional view illustrating the chip package structure along a sectional line **II-II'** in FIG. **13B-1**, in accordance with some embodiments.

(120) After the step of FIG. **1A**, as shown in FIGS. **13A** and **13A-1**, an adhesive layer **174** is formed over the chip structure **132**, in accordance with some embodiments. The adhesive layer **174** is made of a combination of polymer and metal (e.g., a silver paste) or a polymer (e.g., epoxy or silicone), in accordance with some embodiments.

(121) Thereafter, as shown in FIGS. **13B**, **13B-1**, and **13B-2**, a heat-spreading lid **1310** is bonded to the adhesive layers **150** and **174**, in accordance with some embodiments. In this step, a chip package structure **1300** is substantially formed, in accordance with some embodiments. The heat-spreading lid **1310** covers the chip structure **132** and the devices **140**, in accordance with some embodiments. The heat-spreading lid **1310** is formed in one piece, in accordance with some embodiments.

(122) The heat-spreading lid **1310** has a ring structure **1312** and a top plate **1314**, in accordance with some embodiments. The top plate **1314** is over the ring structure **1312**, the chip structure **132**, the adhesive layers **150** and **174**, and the devices **140**, in accordance with some embodiments. The top plate **1314** has openings **1314a** and **1314b**, in accordance with some embodiments. The opening **1314a** is between the chip structure **132** and the ring structure **1312**, in accordance with some embodiments. The openings **1314b** respectively expose the chip structures **122** and **122'**, in accordance with some embodiments.

(123) The heat-spreading lid **1310** is made of a rigid material, such as metal (e.g., copper or iron), alloys thereof (e.g., stainless steel), or another suitable material which is more rigid than the wiring substrate **110**, in accordance with some embodiments.

(124) Processes and materials for forming the chip package structures **200**, **300**, **400**, **500**, **600**, **700**, **800**, **900**, **1000**, **1100**, **1200**, and **1300** may be similar to, or the same as, those for forming the chip package structure **100** described above.

(125) In accordance with some embodiments, chip package structures and methods for forming the same are provided. The methods (for forming the chip package structure) form an opening in a top plate of a heat-spreading lid. The heat-spreading lid covers a chip structure. The chip structure has a first sidewall closer to a ring structure of the heat-spreading lid than other sidewalls of the chip structure, and the opening is between the first sidewall and the ring structure. The opening is able to hinder the transmission of the thermal stress from the first sidewall to the ring structure through the top plate. Therefore, the opening is able to reduce the thermal stress in the ring structure close to the first sidewall, and therefore the opening is able to prevent an adhesive layer over (or under) the ring structure close to the first sidewall from damage in subsequent thermal processes.

(126) In accordance with some embodiments, a chip package structure is provided. The chip package structure includes a wiring substrate. The chip package structure includes a first chip structure over the wiring substrate. The chip package structure includes a heat-spreading lid over the wiring substrate and covering the first chip structure. The heat-spreading lid includes a ring structure and a top plate. The ring structure surrounds the first chip structure. The top plate covers the ring structure and the first chip structure. The first chip structure has a first sidewall and a second sidewall opposite to the first sidewall, a first distance between the first sidewall and the ring structure is less than a second distance between the second sidewall and the ring structure, the top plate has a first opening, the first opening has a first inner wall and a second inner wall facing each other, and the first inner wall and the second inner wall are between the first sidewall and the ring structure in a top view of the heat-spreading lid and the first chip structure.

(127) In accordance with some embodiments, a chip package structure is provided. The chip package structure includes a wiring substrate. The chip package structure includes a first chip structure and a second chip structure over the wiring substrate and spaced apart from each other by a gap. The chip package structure includes a heat-spreading lid over the wiring substrate and covering the first chip structure. The heat-spreading lid includes a ring structure and a top plate, the ring structure surrounds the first chip structure and the second chip structure, the top plate covers the ring structure and the first chip structure, the ring structure has a first portion and a second portion, the first chip structure and the second chip structure are between the first portion and the second portion, the first portion is closer to the first chip structure than the second portion, and the top plate has a first opening between the gap and the first portion.

(128) In accordance with some embodiments, a method for forming a chip package structure is provided. The method includes disposing a first chip structure over a wiring substrate. The method includes forming a heat-spreading lid over the wiring substrate. The heat-spreading lid covers the first chip structure, the heat-spreading lid comprises a ring structure and a top plate, the ring structure surrounds the first chip structure, the top plate covers the ring structure and the first chip structure, the first chip structure has a first sidewall and a second sidewall opposite to each other, the first sidewall is closer to the ring structure than the second sidewall, the top plate has a first opening between the first sidewall and the ring structure, the first opening has a first inner wall and a second inner wall facing each other, and the second inner wall is between the first inner wall and the first sidewall in a top view of the heat-spreading lid and the first chip structure.

(129) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A chip package structure, comprising: a wiring substrate; a first chip structure over the wiring substrate; and a heat-spreading lid over the wiring substrate and covering the first chip structure, wherein the heat-spreading lid comprises a ring structure and a top plate, the ring structure surrounds the first chip structure, the top plate covers the ring structure and the first chip structure, the first chip structure has a first sidewall and a second sidewall opposite to the first sidewall, the first sidewall is closer to the ring structure than the second sidewall, the top plate has a first opening, the first opening is between the first sidewall and the ring structure in a top view of the heat-spreading lid and the first chip structure, the ring structure has a second opening, the first chip

structure is in the second opening, the second opening has a first inner wall facing the first chip structure, the first inner wall has a recess, the first opening overlaps the recess in the top view, and the recess of the first inner wall of the second opening of the ring structure is closer to the wiring substrate than the first opening of the top plate.

2. The chip package structure as claimed in claim 1, wherein the first chip structure has a third sidewall connected between the first sidewall and the second sidewall, the first chip structure has a first corner between the first sidewall and the third sidewall, and the first opening is between the first corner and the ring structure.

3. The chip package structure as claimed in claim 2, wherein the first chip structure has a fourth sidewall opposite to the third sidewall, the first chip structure has a second corner between the first sidewall and the fourth sidewall, and the first opening is further between the second corner and the ring structure.

4. The chip package structure as claimed in claim 2, wherein the first chip structure has a fourth sidewall adjacent to the first sidewall and the second sidewall, the first chip structure has a second corner between the first sidewall and the fourth sidewall, the top plate further has a second opening between the second corner and the ring structure, and the second opening is spaced apart from the first opening.

5. The chip package structure as claimed in claim 1, further comprising: an adhesive layer between the ring structure and the top plate.

6. The chip package structure as claimed in claim 1, wherein the first chip structure has a first corner adjacent to the first sidewall, the first opening has a first wide portion and a narrow portion, the first wide portion is wider than the narrow portion, and the first wide portion is between the first corner and the ring structure.

7. The chip package structure as claimed in claim 6, wherein the first chip structure further has a second corner adjacent to the first sidewall, the first opening further has a second wide portion, the second wide portion is wider than the narrow portion, and the second wide portion is closer to the second corner than the narrow portion.

8. The chip package structure as claimed in claim 1, wherein the recess of the first inner wall of the second opening of the ring structure is wider than the first opening of the top plate.

9. A chip package structure, comprising: a wiring substrate; a first chip structure and a second chip structure over the wiring substrate and spaced apart from each other by a gap; and a heat-spreading lid over the wiring substrate and covering the first chip structure, wherein the heat-spreading lid comprises a ring structure and a top plate, the ring structure surrounds the first chip structure and the second chip structure, the top plate covers the ring structure and the first chip structure, the ring structure has a first portion, a second portion, and a third portion, the first chip structure is between the first portion and the second portion, the first portion is closer to the first chip structure than the second portion, the second chip structure is between the third portion and the second portion, a first line width of the first portion is less than a second line width of the third portion, the first portion and the third portion are made of a same material, and the top plate has a first opening between the gap and the first portion.

10. The chip package structure as claimed in claim 9, wherein the gap extends toward the first opening in a top view of the heat-spreading lid, the first chip structure, and the second chip structure.

11. The chip package structure as claimed in claim 9, wherein the first chip structure has a corner adjacent to the gap, and the first opening surrounds the corner.

12. The chip package structure as claimed in claim 9, wherein the first line width of the first portion is less than a third line width of the second portion.

13. The chip package structure as claimed in claim 12, wherein the ring structure further has a fourth portion, the first portion is connected between the third portion and the fourth portion, and the first line width of the first portion is less than a fourth line width of the fourth portion.

14. The chip package structure as claimed in claim 13, further comprising: a third chip structure over the wiring substrate and between the fourth portion and the second portion of the ring structure.

15. A chip package structure, comprising: a wiring substrate; a first chip structure over the wiring substrate; and a heat-spreading lid over the wiring substrate, wherein the heat-spreading lid covers the first chip structure, the heat-spreading lid comprises a ring structure and a top plate, the ring structure surrounds the first chip structure, the top plate covers the ring structure and the first chip structure, the first chip structure has a first sidewall, a second sidewall, and a third sidewall connected between the first sidewall and the second sidewall, the third sidewall faces a first portion of the ring structure, the first sidewall is closer to the ring structure than the second sidewall, a first distance between the third sidewall and the first portion of the ring structure is greater than a second distance between the first sidewall and the ring structure and greater than a third distance between the second sidewall and the ring structure, the first chip structure has a first corner between the first sidewall and the third sidewall, the top plate has a first opening between the first corner and the ring structure in a top view of the heat-spreading lid and the first chip structure.

16. The chip package structure as claimed in claim 9, wherein the second portion of the ring structure is made of the material.

17. The chip package structure as claimed in claim 15, wherein the first chip structure has a fourth sidewall opposite to the third sidewall, the first chip structure has a second corner between the first sidewall and the fourth sidewall, and the first opening is further between the second corner and the ring structure.

18. The chip package structure as claimed in claim 15, wherein the first chip structure has a fourth sidewall adjacent to the first sidewall and the second sidewall, the first chip structure has a second corner between the first sidewall and the fourth sidewall, the top plate further has a second opening between the second corner and the ring structure, and the second opening is spaced apart from the first opening.

19. The chip package structure as claimed in claim 15, wherein the first opening has a U-shape.

20. The chip package structure as claimed in claim 18, wherein the fourth sidewall faces a second portion of the ring structure, a fourth distance between the fourth sidewall and the second portion of the ring structure is greater than the second distance between the first sidewall and the ring structure and greater than the third distance between the second sidewall and the ring structure.
